

Jamie Nagle's toy exercises with OT flow matching

Does flow matching keep the jet, its medium-induced radiation and the event's own underlying event?
Four exercises on a toy whose truth is known

- **Subtraction:** no halo, but the solved flow gives 25% of the jet core to the background, and on quenched jets 71-82% of the recoil. No loss change fixes it: the network's one-step estimate is right; the transport is not
- **Translation:** a deterministic flow keeps each jet and under-quenches; the noise-driven model gets the quenched population, not the jet-by-jet randomness, unless trained on true pairs
- **With the UE:** the matching cost decides. With a near/far cost the flow keeps this event's UE as exactly as the truth and recovers 114% / 80% of the core / ring change
- **Prior against data:** the data change the answer measurably, but the prior still decides most of it

Every result is labelled by its supervision: **pure unpaired** independent pools, paired by the OT plan **synthetic paired** mixtures built from the prior and UE pools, trained to their own UE **hybrid** both **true-paired control** trained on true pairs (a positive control)

TOY MODEL: a reimplement of the deck's toycalo listings, not PYTHIA, HIJING or JEWEL. Seed 0 unless stated; 20k held-out events per test; 32 A6000 GPU hours of training. Full log FLOW_NOTES.md, results docs/flow/jamie_otfm

Setup

The toy (reimplemented from Jamie's listings; his code was not available)

- 24×64 towers, $|\eta| < 1.1$, periodic φ ; jets E^{-4} in 20-80 GeV, one or two prongs; UE: gamma towers (mean 0.6 GeV, v_2) plus minijets
- quenching: $f \sim \text{Beta}(2,6)$; particles lose $(1 - f)$ and spread by $(1 + f)$; fE goes into ≈ 15 recoil particles at $0.3 < r < 1.0$. Each draw is random: one vacuum jet, many quenched versions
- checks against Jamie's numbers: cone $28.4 \rightarrow 22.0$ GeV ($28.2 \rightarrow 21.9$), ring $0.4 \rightarrow 6.6$ ($0.4 \rightarrow 6.6$), ρA offset/RMS $+0.85/4.5$ GeV ($+0.7/4.4$), UE-only fakes 88% (88%)
- separate parents for every pool (40k each) and test set (20k); the hidden truth only scores

Regions around the jet axis: core $R < 0.4$, recoil ring 0.4-1.0, far $R > 1.0$ (towers there change in 22% of quenched jets: the protected region is not exactly untouched)

The models (the UVCGAN-S ViT network, 32M; log-energy coordinates; batch 256)

- **D**: deterministic OT-CFM. A source image flows to its OT-matched target; one output per input
- **C**: conditional FM on the OT pseudo-joint. The input is a fixed second channel; noise flows to the matched target; one image per noise draw
- **the OT plan** (exact, squared distance of whole images) matches the UE in subtraction and Exercise 4 and the position in Exercise 3, hardly the jet's energy (an audit before training). Exercise 4 also runs the pre-specified near/far cost
- 1.5 GPU-hours per baseline; continuations 4000 updates from it with one change; extra-term weights from a gradient-scale audit, frozen
- the ODE: midpoint, NFE doubled on validation until stable, frozen before testing

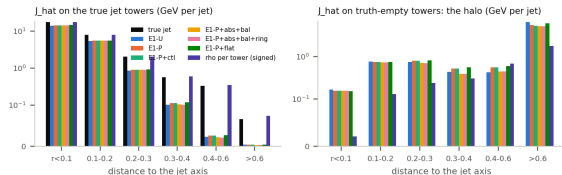
Exercise 1: subtraction. The core goes into the background; no halo

		offset GeV	RMS GeV	RMS cal. GeV	halo $R < 0.4$ GeV, +	hard-tower loss	negative towers	fakes: largest cone > 10	fakes: anti- k_T
E1-U	pure unpaired	-5.6	2.53	2.54	2.1	-26%	28%	0.1%	0.0%
E1-P	synthetic paired	-5.0	2.53	2.62	2.2	-25%	2.7%	0.1%	0.1%
E1-P+ctl	synthetic paired	-5.0	2.56	2.64	2.2	-24%	1.9%	0.1%	0.1%
E1-P+abs+bal	synthetic paired	-5.2	2.55	2.64	2.0	-24%	0.2%	0.1%	0.1%
E1-P+abs+bal+ring	synthetic paired	-5.3	2.57	2.65	2.0	-24%	0.2%	0.1%	0.1%
E1-P+flat	synthetic paired	-4.7	2.64	2.68	2.3	-23%	2.0%	0.0%	0.0%
rho per tower (signed)	standard	0.8	4.54	4.55	9.1	0.3%	61%	—	—
rho per tower, negatives dropped	standard	9.5	3.63	3.64	9.1	0.3%	0.0%	—	—
rho x A (cone)	standard	0.8	4.54	4.55	—	—	—	88%	100%

20k held-out mixtures (jets 20-80 GeV in UE), true axis, $\hat{J} = M - \hat{B}$ raw and signed. Hard-tower loss: $\hat{J} - J$ on true towers above 1 GeV. Halo: positive $\hat{J}$ on truth-empty towers in the cone. Fakes: UE-only events. Continuations: 4000 updates from E1-P, one change each.

- **Jamie's core loss is reproduced, his halo is not** (pre-registered: loss above 10%, halo above 3 GeV), **pure unpaired** and **synthetic paired** alike
- resolution 2.5 GeV against ρA 's 4.5 (Jamie's fixed network: 2.4-2.6); fakes 0.1%
- the offset grows with energy (-4.6 GeV at 10-20, -8.8 at 50-80 for E1-U)
- **what the loss changes:** GeV and balanced masks remove the negative towers (2.7% $\rightarrow$ 0.2%), trim the halo (2.2 $\rightarrow$ 2.0 GeV); ring sums add nothing; the flat prior flattens the energy dependence
- **what it does not:** the core loss (-23 to -25%) and the offset

Exercise 1: why the core is lost



run	estimate	hard-tower loss	cone offset GeV	$\hat{B}$ on hard towers GeV (true UE 0.58)
E1-P_s0	one-step mean at $t = 0$	-2.2%	1.7	0.71
E1-P_s0	solved, 32 NFE	-25%	-5.2	1.98
E1-U_s0	one-step mean at $t = 0$	1.0%	7.3	0.53
E1-U_s0	solved, 32 NFE	-26%	-5.8	2.07
E1-P_absbal	one-step mean at $t = 0$	-1.8%	0.9	0.69
E1-P_absbal	solved, 32 NFE	-24%	-5.4	1.96

One-step mean: $x_0 + v(0, x_0)$, the regression's estimate of the mean background given the mixture; solved: the ODE output that is scored. 500 validation mixtures.

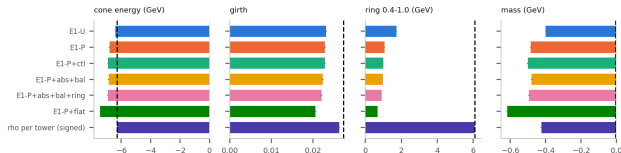
- ring by ring (Jamie's table): the true jet towers keep 14.2-14.4 of 17.7 GeV at $r < 0.1$ and lose most of their soft edge; empty towers get under 1 GeV per ring within $R < 0.6$ (4-6 GeV spread over the far towers)
- the network's one-step estimate has the core right (hard-tower error -2% , $+1\%$; $\hat{B}$ 0.5-0.7 GeV on hard towers, true UE 0.58) and shows Jamie's halo instead (UE fluctuations left in the jet: cone offset $+1.7$, $+7.3$ GeV)
- the solved flow loses 25%: $\hat{B}$ is 2.0 GeV on the hard towers. The flow carries the mixtures onto realistic UE images, and puts the UE's largest fluctuations where the mixture is brightest: under the jet core
- so losses on the one-step surrogate (Jamie's three-part fix, as specified) cannot reach it: their terms barely fall in training (abs 0.233 $\rightarrow$ 0.216)

Exercise 2: a frozen vacuum subtractor on quenched jets

		cone E shift	girth shift	ring 0.4-1 shift	mass shift	quenched jets lost	recoil in $\hat{B}$	ring of vacuum jets
truth		-6.27 GeV	+0.027	+6.09 GeV	-0.005 GeV	0	–	0.40 GeV
E1-U	pure unpaired	102%	85%	29%	-0.40 GeV	14%	4.35	1.50
E1-P	synthetic paired	108%	84%	18%	-0.48 GeV	15%	5.01	1.65
E1-P+ctl	synthetic paired	110%	84%	16%	-0.50 GeV	16%	5.10	1.61
E1-P+abs+bal	synthetic paired	109%	82%	16%	-0.48 GeV	17%	5.12	1.45
E1-P+abs+bal+ring	synthetic paired	110%	81%	15%	-0.49 GeV	18%	5.18	1.44
E1-P+flat	synthetic paired	118%	75%	11%	-0.62 GeV	21%	5.42	1.78
rho per tower (signed)	standard	100%	96%	100%	-0.42 GeV	3.1%	–	2.87

The same 20k jets, vacuum and quenched, over the same UE; quenched minus vacuum, mean per jet; %: of the true shift. Lost: below 10 GeV while the truth is above. Recoil in $\hat{B}$: the ring change that went into the background, GeV.

Exercise 2: a frozen subtractor on the same jets, vacuum and quenched, over the same UE (20k); quenched - vacuum, mean per jet; dashed: truth



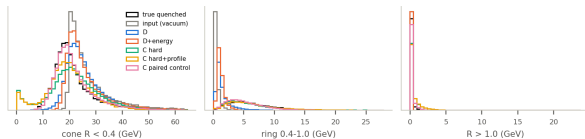
- the recoil goes into the background: 4.4-5.4 of the 6.1 GeV ring change ends in $\hat{B}$ (Jamie: 98%)
- core change about right (102-118%; Jamie 136%), girth 75-85% (69%)
- quenched cores lose more: 40% of the hard-tower energy, 25% in vacuum
- no continuation of Exercise 1 changes it; the flat prior is the worst
- the mass change (-0.4 GeV) appears for the standard readout too: not a clean observable here

Exercise 3: clean translation. Population against correspondence

	W_1/σ to the true quenched				shift recovered			$r(\text{cone, input})$
	cone	ring	girth	$R > 1$	cone	ring	girth	
truth reference (independent sample)	0.010	0.023	0.008	0.015				
identity (input)	0.718	1.365	0.630	0.276	-0.0%	0.0%	0.0%	1.00
D	0.449	1.009	0.407	0.285	39%	26%	36%	0.86
D+ctl	0.362	1.009	0.397	0.275	52%	26%	37%	0.86
D+energy	0.507	1.174	0.483	0.310	29%	14%	23%	0.97
D seed 1	0.357	0.963	0.361	0.278	53%	30%	44%	0.90
C hard	0.295	0.214	0.916	3.396	103%	109%	245%	0.07
C soft	0.368	0.254	1.036	6.242	113%	108%	264%	0.08
C paired control	0.044	0.224	0.096	0.216	95%	84%	85%	0.86
C hard+ctl	0.242	0.187	0.959	3.547	128%	111%	252%	0.07
C hard+profile	0.247	0.147	0.842	3.312	133%	110%	234%	0.09
C hard+div	0.259	0.230	1.010	3.205	133%	116%	260%	0.07

20k held-out vacuum jets, one output each, against their true quenched versions; W_1/σ : population distance (an independent quenched sample: 0.01-0.02). Shift recovered: 100% = right.

Exercise 3: 20k held-out jets, one output each



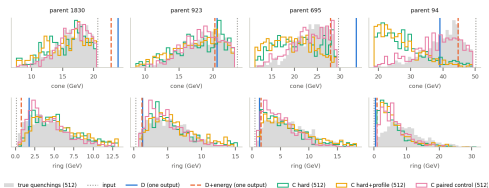
- **D** keeps each jet (r 0.86) and quenches too little: 39-53% of the core shift, 26-30% of the ring. The energy term makes it change less (29%, 14%)
- **C pure unpaired** gets the population means (103-133%) with real recoil towers, but ignores its input (r 0.07) and places jets 0.13 off the axis (the spike at 0): the OT pairs' own offset
- the **true-paired control** control: 95%, 84%, 85%, tied to its input

Exercise 3: the conditional law. Variation is not the right randomness

		conditional W_1 / σ			68% coverage			conditional sd: model / truth		
		cone	ring	girth	cone	ring	girth	cone	ring	girth
truth (true-vs-true reference)		0.065	0.120	0.071	67%	67%	67%	1.00	1.00	0.99
D	pure unpaired	0.645	1.198	0.560	0.0%	0.0%	0.0%	0.00	0.00	0.00
D+ctl	pure unpaired	0.582	1.236	0.517	0.0%	0.0%	0.0%	0.00	0.00	0.00
D+energy	pure unpaired	0.589	1.175	0.514	0.0%	0.0%	0.0%	0.00	0.00	0.00
D seed 1	pure unpaired	0.593	1.175	0.580	0.0%	0.0%	0.0%	0.00	0.00	0.00
C hard	pure unpaired	0.899	0.588	1.109	84%	72%	61%	3.07	1.72	2.84
C soft	pure unpaired	0.973	0.600	1.206	87%	75%	58%	3.31	1.78	2.93
C paired control	true-paired control	0.117	0.282	0.204	72%	60%	68%	1.12	0.89	1.05
C hard+ctl	pure unpaired	0.870	0.575	1.134	82%	71%	59%	2.91	1.66	2.81
C hard+profile	pure unpaired	0.816	0.544	1.044	79%	69%	63%	2.64	1.52	2.74
C hard+div	pure unpaired	0.877	0.572	1.174	82%	67%	54%	2.88	1.56	2.78

100 test parents; 128 model draws each against 128 true quenchings (another 128: the true-vs-true floor). Pre-registered: the law is recovered if $W_1 \leq 1.5 \times$ the paired control's and 68% coverage is 58-78%.

Exercise 3: one input jet, many true quenchings against many model draws (the conditional law)



- **true-paired control**: the network can learn the toy's randomness (coverage 60-72%, widths 0.9-1.1)
- **pure unpaired** C: a parent's draws spread like the whole population (cone sd 10-13 GeV against 3.8): variation, not the law. Profile and diversity terms change nothing
- **girth rule**: the loss is predictable from the input (ceiling r 0.97); C's follows the energy (+0.30), not the girth (-0.09), per jet -0.10. Jamie: +0.32, -0.12, -0.11

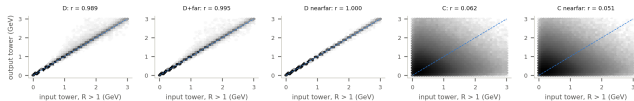
Exercise 4: translate, and keep this event's UE

		far towers: r (output, input)	far rms change / UE tower sd	core shift recovered	ring shift recovered	oracle ring GeV
truth ($J_{\text{med}} + B_a$)		1.000	0.023	100%	100%	6.56
D	pure unpaired	0.989	0.154	148%	23%	3.32
D+ctl	pure unpaired	0.989	0.154	141%	24%	3.48
D+global	pure unpaired	0.995	0.107	116%	15%	2.57
D+far	pure unpaired	0.995	0.108	134%	26%	3.15
D nearfar	pure unpaired	1.000	0.025	114%	80%	5.49
D nearfar+far	pure unpaired	1.000	0.016	108%	68%	4.97
D seed 1	pure unpaired	0.990	0.145	146%	24%	3.27
D seed 1+far	pure unpaired	0.996	0.095	124%	31%	3.00
C	pure unpaired	0.060	1.513	397%	30%	6.15
C+far	pure unpaired	0.254	1.191	408%	5.4%	-12.42
C nearfar	pure unpaired	0.050	1.389	215%	176%	11.14
C nearfar+far	pure unpaired	0.069	1.324	367%	46%	-1.70

- whole-image OT pairs events by their UE: D spreads 6 GeV of the core over it (r 0.989); penalties reach 0.995-0.996, at the 10% line (0.095-0.108; the verdict flips with the seed)
- **the near/far cost keeps the event's UE as the truth does** (0.9997) and recovers 114% of the core change, 80% of the ring (Jamie's best: 84%, 69%)
- C regenerates the UE with every cost and penalty (r 0.05-0.25): its spread is new UE

Input $J_{\text{vac}} + B_a$, truth $J_{\text{med}} + B_a$. Pre-registered: the event's UE is kept if far towers correlate at $r \geq 0.99$ and change by $\leq 10\%$ of the UE tower sd. Shifts with ρ from the image's own $R > 1.2$; oracle: the true UE subtracted.

Exercise 4: the UE far from the jet, tower by tower (2000 events)



Prior against data: the 2×2

		cone E shift	girth shift	ring 0.4-1 shift	mass shift	quenched jets lost	recoil in $\hat{B}$	ring of vacuum jets
truth		-6.27 GeV	+0.027	+6.09 GeV	-0.005 GeV	0	–	0.40 GeV
E1-P	synthetic paired	108%	84%	18%	-0.48 GeV	15%	5.01	1.65
vac prior: syn	synthetic paired	108%	83%	14%	-0.48 GeV	12%	5.26	1.44
vac prior $\times$ vac data	hybrid	109%	76%	11%	-0.51 GeV	17%	5.42	1.36
vac prior $\times$ quench data	hybrid	104%	90%	30%	-0.43 GeV	14%	4.24	2.32
broad prior: syn	synthetic paired	109%	86%	20%	-0.49 GeV	13%	4.84	2.91
broad prior $\times$ vac data	hybrid	110%	80%	13%	-0.51 GeV	17%	5.31	1.90
broad prior $\times$ quench data	hybrid	106%	84%	17%	-0.50 GeV	16%	5.03	3.03
vac $\times$ quench + UE profile	hybrid	114%	63%	-24%	-0.58 GeV	29%	7.54	-3.92
rho per tower (signed)	standard	100%	96%	100%	-0.42 GeV	3.1%	–	2.87

Each cell: 4000 updates from E1-P (a vacuum prior); **hybrid**: synthetic sums from the cell's prior + OT pairs of an independent mixture pool ($\lambda_U = 1$; both reach the gradient). Scored as Exercise 2; bar chart: appendix A7.

prior	quenched minus vacuum data: shift (sd)			ring of vacuum jets above truth, GeV			data matter?
	cone, GeV	girth	ring, GeV	vacuum data	quenched data	broad – vacuum prior	
vacuum	0.28 (6.0)	0.0037 (14.9)	1.18 (71.6)	0.95	1.91	–	yes
broad	0.22 (4.6)	0.0010 (4.1)	0.28 (22.8)	1.49	2.63	0.54, 0.72	yes

- **the data matter** (beyond 3 sd): quenched data raise the recovered recoil 11% $\rightarrow$ 30%, girth 76% $\rightarrow$ 90% (vacuum prior). Jamie: identical to three digits
- **not specifically**: they add ring to vacuum jets too; recoil $\leq$ 30%
- **the prior invents recoil**: broad prior +0.5-0.7 GeV in vacuum jets (Jamie +0.7)
- the UE-profile term breaks the subtraction (-3.9 GeV ring)

A second seed, and an accurate solve

Second seed (the same inputs, both seeds; all **pure unpaired**)

	cone change		ring change		$r(\text{change, true change})$		far towers
	$r(\text{seeds})$	rms diff./rms	$r(\text{seeds})$	rms diff./rms	cone	ring rms diff., GeV	
Ex. 3: D vs D seed 1	0.65	0.73	0.81	0.59	0.30	-0.02	0.000
Ex. 4: D vs D seed 1	0.85	0.27	0.90	0.39	0.08	0.10	0.037
Ex. 4: D+far vs D seed 1+far	0.80	0.32	0.84	0.46	0.11	0.11	0.033

The best any deterministic predictor can reach with one true quenching's change: r 0.48-0.50 (the bank's 256 quenches per parent).

- Exercise 4: seeds agree jet by jet, neither predicts the jet's change; populations agree within 2 points. **Agreement is not correctness** (Jamie: 0.84, 0.93; -0.02)
- Exercise 3 D: population cone shift 39% against 53%

Accurate solve

- all runs resolve at 32-256 NFE (validation) except the clean-jet D family and one penalty run at the 256 cap (1.6%)
- **E3-D does not converge**: starts 10^{-3} apart end $580\times$ further apart, all from towers empty at both ends; 1-1.5 GeV per-jet noise at 256-1024 NFE
- **a constraint learned through a coarse rollout holds for that rollout**:

continuation	term	rollout NFE	value	test NFE	value	ratio
E3-Ch_div	div	8	-0.194	64	-0.175	0.9
E3-Ch_profile	profile	8	0.054	128	0.122	2.3
E3-D_energy	energy	32	0.006	256	0.008	1.4
E4-C_far	inv_far	8	0.091	128	0.587	6.4
E4-Cnf_far	inv_far	32	0.791	32	0.791	1.0
E4-D_far	inv_far	8	0.004	64	0.007	1.8
E4-D_global	inv_global	8	0.005	256	0.018	3.7
E4-D_s1_far	inv_far	16	0.004	64	0.006	1.4
E4-Dnf_far	inv_far	8	0.002	64	0.003	1.3

Answers

1. **Halo, core loss, lost recoil?** Core loss (25%) and lost recoil (71-82% into $\hat{B}$): yes; halo: no (2.1 GeV). Cause: the solve, not the regression. Losses change only the side effects (negative towers, halo -0.2 GeV, energy slope)
2. **Does a vacuum subtractor keep the medium modification?** The core change and most of the widening, not the recoil (18-29%); quenched cores lose 40%
3. **Conditional law or variation?** **pure unpaired** C: variation with the right population, wrong law (output r 0.07 with its input). **true-paired control** C: the law. The unpaired pairs, not the network, are missing
4. **This event's UE, or the population?** D with the near/far cost: this event's (as the truth). D with the whole-image cost: almost. C: the population only, whatever the penalty
5. **Do the mixture data matter beyond the prior?** Measurably (recoil 11% $\rightarrow$ 30%), unlike in Jamie's toy, but not specifically, and the prior still invents recoil
6. **What survives a second seed and an accurate solve?** Exercise 4's conclusions (both seeds, resolved solves); Exercises 1-2 and the C verdicts rest on resolved solves (one seed). Not the clean-jet D numbers (unresolved ODE, 14-point seed spread), and not constraints learned through coarse rollouts

Population agreement, event-preserving correspondence and the conditional law are reported separately throughout. A success on this toy would justify a test on PYTHIA/JEWEL; it would not by itself establish the physical modification.

Limits, and what was not run

Limits

- a reimplemented toy; unlisted details (grid, girth, UE redraws) are our choices
- one seed except the two confirmations; 1.5 GPU hours per baseline, 4000-update continuations; the 2×2 are warm starts from one vacuum-prior checkpoint
- the clean-jet D family is numerically unresolved; rollout terms trained through coarser solves than the test solve (except the last C arm)
- matching costs: whole-image squared distance, and for Exercise 4 the one pre-specified near/far alternative (added for C after its whole-image run; disclosed)

Not run

- PYTHIA jets in toy UE (Exercise 1 extension); our paired two-channel decomposer
- C's diversity term in Exercise 4; a resampled-UE diagnostic
- network widths, a compact backbone; longer training beyond the 4000-update controls
- Jamie's discriminator, learning-rate and cycle-weight changes: no counterpart in OT-CFM

Next, if pursued

- a solve or path that keeps the subtractor's core (e.g. a noise-smoothed path, or reading out the one-step mean with a calibrated UE fluctuation)
- a pairing that carries the jet's energy for C

A1. Coverage: Jamie's changes and their flow-matching counterparts (1/2)

Jamie tried (printed slides)	his result	FM counterpart (runs)	our result
GeV error per tower + balanced groups of jet, near-empty and far towers (Ex. 1) (51)	offset +9.9 -> +3.4 GeV; halo 9.6 -> 4.5 GeV	the same terms on the local endpoint surrogate ($t \leq 0.25$), continuation of E1-P (E1-P_absbal (B2) vs E1-P_ctl (B1))	no halo to fix (2.2 GeV); core loss unchanged (-24%); negative towers 2.7% -> 0.2%; halo 2.2 -> 2.0 GeV. The terms act on the one-step surrogate, which is already right; the core is lost in the solve
+ ring energy sums (Ex. 1) (51-52)	offset +1.0 GeV, halo 3.4, scatter 2.6 GeV, fakes 0.3%	+ ring sums, same edges (surrogate) (E1-P_absbalring (B3))	no further change (core -24%, halo 2.0 GeV, offset -5.3 GeV)
radial-profile rule, averaged over jets (Ex. 3) (90)	shift recovered core 95%, ring 80%, girth 36%	batch-mean radial profile of C's solved outputs against an independent quenched batch (E3-Ch_profile (B8) vs E3-Ch_ctl (B7))	population ring W1 0.21 -> 0.15, cone over-quenched (133%); conditional law still wrong (cone W1 0.82); the term holds only partly at the test solve (x2.3)
total energy out = in (Ex. 3 v9) (85-87; 145)	stops the lazy solutions; core 89%, ring 19%, girth 32%; energy 96% +/- 19% per jet	total-GeV term on a differentiable 16-image midpoint rollout, continuation of E3-D (E3-D_energy (B6) vs E3-D_ctl (B5))	pushes D toward changing less: core 39% -> 29%, ring 26% -> 14%, more haze; the toy itself keeps 99.72% of the energy (exact equality is a prior)
dice: random numbers fed to the generator (Ex. 3 v10-v11, Ex. 4 v5) (91-92; 145-146)	samples 1/3 of the true spread (Ex. 3), then ignored with longer training; ignored (Ex. 4)	C: noise-driven conditional FM on the OT pseudo-joint, hard and entropic plans; true-paired control (E3-Ch, E3-Cs, E3-Cp; E3-g-C; E4-C)	C makes real variation but not the conditional law: a parent's spread is the population's (cone sd 11.6 vs 3.8 GeV), output r 0.07 with the input; the true-paired control recovers the law (coverage 60-72%)
diversity rule: different dice, different output (Ex. 3 v12, Ex. 4 v6) (92; 145-146)	spread a free knob: 1.6 vs 4.7 GeV (Ex. 3), 8.0 vs 4.5 GeV (Ex. 4)	capped mode-seeking term between two solved C draws (cap from target pairs) (E3-Ch_div (B9); Ex. 4: not run (budget))	changes nothing (C already too diverse; cone W1 0.88); Ex. 4 not run
no identity rule (Ex. 4 v1) (98; 146)	half the ring, core overshoot, UE inverted (far-tower r -0.52)	D without any invariance term; its unchanged continuation (E4-D (A9), E4-D_ctl (B10))	D keeps the UE nearly (far r 0.989, no inversion) but spreads 6 GeV of the core over it; C regenerates the UE (r 0.06)
identity rule everywhere (Ex. 4 v2) (99; 146)	near-copy: UE kept, no ring	global output-change penalty on solved endpoints (E4-D_global (B11))	far r 0.995, rms 0.108 (still above the 10% limit); suppresses the quenching (ring 15%)
identity rule only at $R > 1.0$, + energy conservation (Ex. 4 v3-v5) (99-100; 146)	near-copy at first (core 30%, no ring); after 4x training core 78-84%, ring 65-70%, UE unchanged	output-change penalty at $R \geq 1.0$ on solved endpoints; also with the near/far matching cost and for C (E4-D_far (B12), E4-Dnf_far (B13), E4-C_far (B15))	D: far r 0.995, ring 26% (full cost); with the near/far cost 0.9999, core 108%, ring 68%; C: far r 0.25 (8-NFE rollout, holds only at 8 NFE) and 0.07 (accurate rollout)
jets below the 20 GeV training range; jets flat in 5-80 GeV (Ex. 1, v7) (54; 145)	-6 GeV at 5-10 GeV: bent toward known jets; flat 5-80 GeV: better above 60 GeV, 5-15 GeV still poor	5-20, 60-80, 80-100 GeV test slices; flat 5-80 GeV prior continuation (E1-U, E1-P; E1-P_flat (B4))	slices: -3.1/-9.6/-10.8 GeV (E1-U at 5-20/60-80/80-100); flat prior: slope 0.94 -> 0.98, 60-80 GeV -6.9 -> -5.7 GeV, 5-20 GeV worse (-4.0), core loss unchanged

A1. Coverage (2/2)

Jamie tried (printed slides)	his result	FM counterpart (runs)	our result
longer training (Ex. 1 v8: x4; Ex. 3 v11: x4, +2400 steps; Ex. 4 v4: x4) (50-51; 90; 145-146) two trainings, different random start (Ex. 4 v4s2) (101; 146)	Ex. 1 offset +13 -> +11 GeV; Ex. 3 core 78% -> 155%; Ex. 4 near-copy -> core 78%, ring 65% agree jet by jet (r 0.84, 0.93), not with the truth (-0.02)	unchanged-loss continuations of 4000 updates; validation curves (E1-P_ctl, E3-D_ctl, E3-Ch_ctl, E4-D_ctl) seed 1 of E4-D (+ far penalty) and of E3-D; same inputs' outputs compared (E4-D_s1 (A13), E4-D_s1_far (B14), E3-D_s1 (A12))	+4000 updates: Ex. 1 unchanged; Ex. 3 D cone 39% -> 52%; C, Ex. 4 D unchanged
loss set by the jet's girth (Ex. 3 toy variant) (109) vacuum or broad prior x vacuum or quenched data (Ex. 1 setup) (116-119)	losses follow energy (+0.32) not width (-0.12); per jet vs truth -0.11 vacuum prior: girth +0.019/+0.020, ring +0.2/+0.2 of +5.7 GeV either data; broad: girth +0.023, ring +0.9 in vacuum jets	D and C trained on separate girth-rule pools (E3g-D (A7); E3g-C (A8)) hybrid L_syn + L_unp (lambda_U = 1), four cells and two synthetic-only arms, continuations of E1-P (PD_vac_vac, PD_vac_q, PD_broad_vac, PD_broad_q (B16-B19); PD_vac_syn, PD_broad_syn (B20-B21))	D: loss follows girth 0.22, energy 0.15, true loss 0.17; C: girth -0.09, energy +0.30, true loss -0.10 (ceiling 0.97) data matter (ring +1.18 GeV, 72 sd, vacuum prior) but recoil <= 30% and quenched data add ring to vacuum jets too; broad prior +0.5-0.7 GeV ring in vacuum jets
batch-averaged UE around the jet must be flat (Ex. 1 setup) (122-123)	girth 61% -> 70%, recoil 0.3 of 5.7 GeV, +2.8 GeV ring in vacuum jets	batch-mean UE radial profile of B_hat against independent UE under the same geometry, on the vacuum-prior x quenched-data cell (PD_vac_q_ueprof (B22))	breaks the subtraction: vacuum-jet ring -3.9 GeV, quenched ring change -24%, 29% of quenched jets lost
vacuum-trained subtractor on quenched jets (Ex. 2) (70-76)	cone 136%, girth 69%, ring 2%, mass -0.7 GeV invented, 12% of quenched jets lost	the frozen Exercise 1 checkpoints on held-out pairs with shared UE (E1-U, E1-P and the E1-P continuations)	cone 102-108%, girth 85%, ring 18-29% (recoil 4.4-5.0 GeV in B_hat), mass -0.4 GeV (as for rho readouts), 14-15% of quenched jets below 10 GeV; continuations change none of it
0.2M -> 0.9M -> 3.5M parameters (Ex. 1) (50) identity term and cycle weight 100; whole-image discriminator heads, three discriminators, cycle weight 10 -> 2 and discriminator learning rate x2 (48; 57-60; 145)	offset +13.4 -> +8.6 GeV identity + cycle 100: halo worse	not run: one fixed backbone (UVCAN-S ViT-ModNet, 32M); no architecture search not applicable: OT-CFM has no discriminator, cycle or identity loss	- -
PYTHIA jets in toy UE (Ex. 1) (55-56) - (-)	the same halo and the same cure -	deferred (budget) our paired two-channel decomposer on the toy: optional, not run	- -

A2. Run ledger (training time on A6000s; 32.4 GPU hours in all)

run	model	supervision	from	updates	GPU h	upd./s	NFE
E1-P_absbal	D	synthetic paired	E1-P_s0	4.0k	0.50	2.2	32
E1-P_absbalring	D	synthetic paired	E1-P_s0	4.0k	0.49	2.2	32
E1-P_ctl	D	synthetic paired	E1-P_s0	4.0k	0.25	4.4	32
E1-P_flat	D	synthetic paired	E1-P_s0	4.0k	0.25	4.5	32
E1-P_s0	D	synthetic paired	scratch	24.4k	1.50	4.5	32
E1-U_s0	D	pure unpaired	scratch	23.8k	1.50	4.4	32
E3-Ch_ctl	C	pure unpaired	E3-Ch_s0	4.0k	0.26	4.4	128
E3-Ch_div	C	pure unpaired	E3-Ch_s0	4.0k	0.67	1.7	64
E3-Ch_profile	C	pure unpaired	E3-Ch_s0	4.0k	0.46	2.4	128
E3-Ch_s0	C	pure unpaired	scratch	24.1k	1.50	4.5	64
E3-Cp_s0	C	true-paired control	scratch	24.7k	1.50	4.6	64
E3-Cs_s0	C	pure unpaired	scratch	23.6k	1.50	4.4	128
E3-D_ctl	D	pure unpaired	E3-D_s0	4.0k	0.25	4.4	256
E3-D_energy	D	pure unpaired	E3-D_s0	4.0k	1.21	0.9	256
E3-D_s0	D	pure unpaired	scratch	24.2k	1.50	4.5	256
E3-D_s1	D	pure unpaired	scratch	24.1k	1.50	4.5	256
E3g-C_s0	C	pure unpaired	scratch	24.1k	1.50	4.5	128
E3g-D_s0	D	pure unpaired	scratch	23.9k	1.50	4.4	256
E4-C_far	C	pure unpaired	E4-C_s0	4.0k	0.46	2.4	128

run	model	supervision	from	updates	GPU h	upd./s	NFE
E4-C_s0	C	pure unpaired	scratch	23.6k	1.50	4.4	32
E4-Cnf_far	C	pure unpaired	E4-Cnf_s0	4.0k	1.13	1.0	32
E4-Cnf_s0	C	pure unpaired	scratch	23.1k	1.50	4.3	32
E4-D_ctl	D	pure unpaired	E4-D_s0	4.0k	0.26	4.3	128
E4-D_far	D	pure unpaired	E4-D_s0	4.0k	0.45	2.4	64
E4-D_global	D	pure unpaired	E4-D_s0	4.0k	0.48	2.3	256
E4-D_s0	D	pure unpaired	scratch	23.7k	1.50	4.4	64
E4-D_s1	D	pure unpaired	scratch	23.6k	1.50	4.4	64
E4-D_s1_far	D	pure unpaired	E4-D_s1	4.0k	0.67	1.7	64
E4-Dnf_far	D	pure unpaired	E4-Dnf_s0	4.0k	0.47	2.4	64
E4-Dnf_s0	D	pure unpaired	scratch	23.1k	1.50	4.3	64
PD_broad_q	D	hybrid	E1-P_s0	4.0k	0.50	2.2	64
PD_broad_syn	D	synthetic paired	E1-P_s0	4.0k	0.25	4.4	32
PD_broad_vac	D	hybrid	E1-P_s0	4.0k	0.50	2.2	64
PD_vac_q	D	hybrid	E1-P_s0	4.0k	0.50	2.2	64
PD_vac_q_ueprof	D	hybrid	E1-P_s0	4.0k	0.63	1.8	64
PD_vac_syn	D	synthetic paired	E1-P_s0	4.0k	0.25	4.5	32
PD_vac_vac	D	hybrid	E1-P_s0	4.0k	0.50	2.2	64

From: the base of a continuation (4000 updates); NFE: the frozen midpoint solve of the scored outputs. E3-D_energy, E3-Ch_div, E4-Cnf_far: the rollout terms make their updates slower.

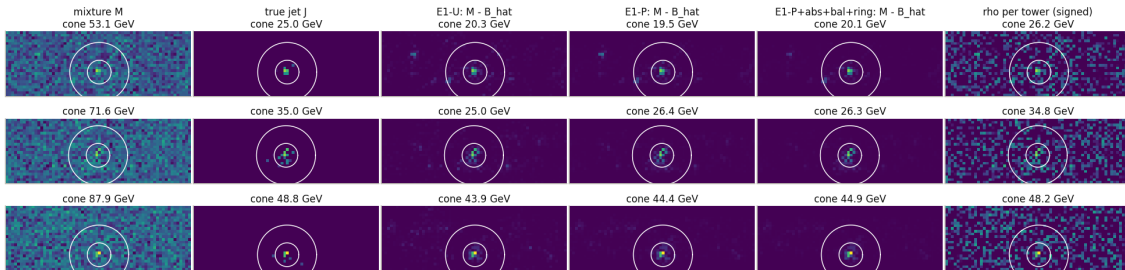
A3. Solver: the frozen accuracy of every baseline

run	frozen		N against 2N (validation)			4 Euler steps	
	NFE	resolved	key / change	W ₁ shifts	key / two noises	key / change	ms / image
E1-U_s0	32	yes	0.001	within sd	—	0.17	1.1 / 8.7
E1-P_s0	32	yes	0.001	within sd	—	0.04	1.1 / 8.7
E3-D_s0	256	no	0.256	beyond sd	—	3.16	1.1 / 70.1
E3-Ch_s0	64	yes	0.009	within sd	0.009	—	—
E3-Cs_s0	128	yes	0.004	within sd	0.004	—	—
E3-Cp_s0	64	yes	0.006	within sd	0.008	0.45	1.1 / 17.6
E3g-C_s0	128	yes	0.003	within sd	0.002	—	—
E4-D_s0	64	yes	0.005	within sd	—	1.19	1.1 / 17.6
E4-D_s1	64	yes	0.009	within sd	—	—	—
E4-Dnf_s0	64	yes	0.003	within sd	—	—	—
E4-C_s0	32	yes	0.001	within sd	0.003	—	—

N against $2N$ on 1000 validation inputs (for C the same noise): adequate when the key per-image energy moves by less than 1% of the model's change, every W_1 shift stays within its bootstrap sd, and for C the change is below 1/10 of that between two noises; otherwise N doubles, to 256 at most. Four Euler steps: a speed diagnostic only. Continuations: checked at their base's N and re-solved by the same rule when they fail.

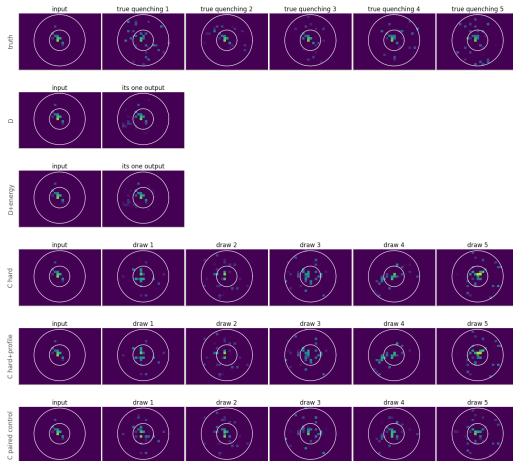
A4. Exercise 1: fixed held-out events

Exercise 1: fixed held-out events (true cone energy 25, 35, 50 GeV); $\log_{10}(E + 0.1)$, circles $R = 0.4$ and 1.0



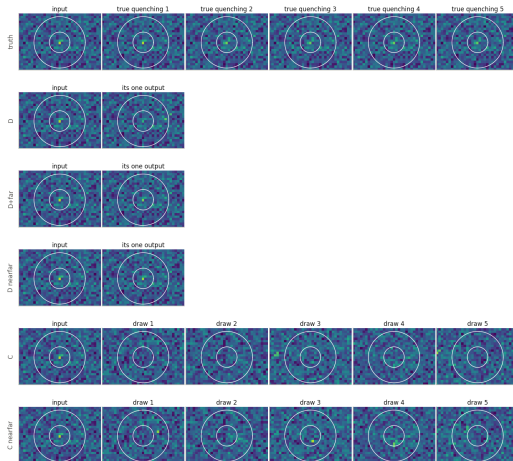
A5. Exercise 3: one parent, true quenchings and model draws

Exercise 3: one fixed parent (cone 23.6 GeV); each panel one image, no averaging; $|\phi - \phi_{\text{jet}}| < 1.6$, circles $R = 0.4, 1.0$



A6. Exercise 4: one parent, true quenchings and model draws

Exercise 4: one fixed parent (cone 24.1 GeV); each panel one image, no averaging; $|\phi - \phi_{\text{jet}}| < 1.6$, circles $R = 0.4, 1.0$



A7. Prior against data: the shifts of every cell

Prior against data: the same 20k pairs, each cell a continuation of E1-P; quenched - vacuum, mean per jet; dashed: truth

